

institutes		
uuid	id	PK
varchar	name	
varchar	slug	UK

institute_id



materials			
uuid	id	PK	
uuid	institute_id	FK	
uuid	subject_id	FK	refs academic structure figure
uuid	chapter_id	FK	refs academic structure figure
uuid	topic_id	FK	refs academic structure figure
varchar	material_type		PDF IMAGE DOCUMENT TEXT
varchar	processing_status		UPLOADED QUEUED PROCESSING READY FAILED
varchar	storage_key		
text	text_content		
int	revision		

material_id



material_enhancements			
uuid	id	PK	
uuid	material_id	FK	
varchar	status		enhanced unchanged
varchar	source_fingerprint	UK	
int	version		
jsonb	payload		

enhancement_id



material_enhancement_segments			
uuid	id	PK	
uuid	enhancement_id	FK	
varchar	level		chapter section other
varchar	relevance		relevant uncertain irrelevant unmapped
text	content		

segment_id



material_enhancement_segment_mappings			
uuid	id	PK	
uuid	segment_id	FK	
uuid	syllabus_topic_id	FK	mapped entity
float	confidence		